This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

RE . .IVITIES: ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES: MC, YD

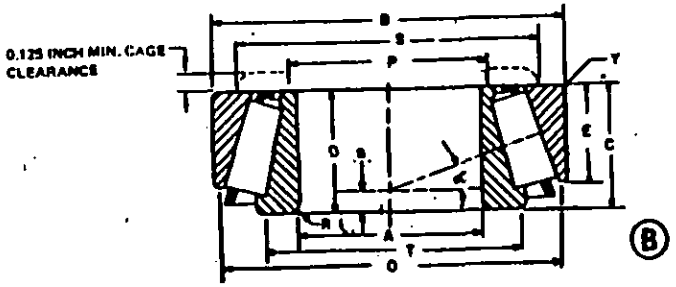

Table I

Dash No.	AFBMAⒷ bearing number (See Note 12)	A	B	C	D	E	R	Y	T	P	S	O	K factor	Basic dynamic load ratings (lb)Ⓑ		Ⓑ Effective load center
	Cone–Cup	Bore	Outside dia.	Bearing width	Cone width	Cup width	Ⓑ Max. shaft fillet radius	Ⓑ Max. housing fillet radius	ⒷRecommended shoulder diameter Shaft		Housing			Radial	Thrust	
1	A4050–A4138	0.5000	1.3775	0.4330	0.4326	0.3437	0.05	0.05	0.73	0.67	1.14	1.26	1.29	500	387	-0.10
2	L21549–L21511	0.6250	1.3775	0.4330	0.4330	0.3430	0.05	0.05	0.85	0.77	1.14	1.28	1.83	575	314	-0.13
•3	A6062–A6162	0.6250	1.6250	0.4687	0.4391	0.3437	0.05	0.05	0.87	0.81	1.36	1.46	1.11	530	480	-0.06
4	17580–17520	0.6250	1.6875	0.6563	0.6563	0.5313	0.06	0.06	0.91	0.83	1.44	1.54	1.76	1,150	655	-0.23
5	05062–05185	0.6250	1.8504	0.5662	0.5662	0.4375	0.06	0.05	0.93	0.83	1.59	1.67	1.64	1,020	620	-0.15
•6	09062–09195	0.6250	1.9380	0.7813	0.8480	0.5625	0.03	0.05	0.87	0.85	1.65	1.75	2.20	1,600	730	-0.36
7	A6075–A6157	0.7500	1.5745	0.4730	0.4391	0.3750	0.04	0.05	0.94	0.91	1.34	1.46	1.11	530	480	-0.06
•8	4A–6	0.7500	1.7500	0.5000	0.4688	0.3750	0.06	0.06	1.00	0.93	1.50	1.61	1.22	815	670	-0.07
9	LM11949–LM11910	0.7500	1.7810	0.6100	0.6550	0.4750	0.05	0.05	0.98	0.93	1.56	1.63	1.94	1,230	635	-0.22
10	05075–05185	0.7500	1.8504	0.5662	0.5662	0.4375	0.05	0.05	0.98	0.93	1.59	1.67	1.64	1,020	620	-0.15
11	09067–09195	0.7500	1.9380	0.7100	0.7500	0.5625	0.05	0.05	1.00	0.94	1.65	1.75	2.20	1,600	730	-0.29
183	09074–09194	0.7500	1.9380	0.9063	0.8480	0.6875	0.06 x 45°	0.14	1.02	0.94	1.54	1.75	2.20	1,600	730	-0.36
•12	1775–1729	0.7500	2.2400	0.7625	0.7810	0.6250	0.06	0.05	1.06	0.98	1.93	2.01	1.90	1,740	915	-0.27
13	12580–12520	0.8125	1.9380	0.7813	0.7813	0.6250	0.06	0.06	1.12	1.02	1.67	1.79	1.81	1,660	915	-0.28
14	M12649–M12610	0.8437	1.9687	0.6900	0.7200	0.5500	0.05	0.05	1.08	1.00	1.73	1.81	2.10	1,660	790	-0.25
15	07087–07196	0.8750	1.9687	0.5313	0.5614	0.3750	0.05	0.04	1.12	1.06	1.75	1.85	1.45	1,130	780	-0.11
16	1755–1729	0.8750	2.2400	0.7625	0.7810	0.6250	0.05	0.05	1.14	1.08	1.93	2.01	1.90	1,740	915	-0.27
17	1779–1729	0.9375	2.2400	0.7625	0.7810	0.6250	0.03	0.05	1.16	1.12	1.93	2.01	1.90	1,740	915	-0.27
18	07100–07196	1.0000	1.9687	0.5313	0.5614	0.3750	0.04	0.04	1.20	1.16	1.75	1.85	1.45	1,130	780	-0.11
184	L44643–L44610	1.0000	1.9800	0.5600	0.5800	0.4200	0.05	0.05	1.24	1.16	1.75	1.85	1.56	1,200	770	-0.13

Ⓑ Denotes changes. For NOTES see sheets 11 and 12.

APPROVED 30 April 1959 REVISED Ⓐ 6 February 1968 Ⓑ 20 August 1976

FED. SUP CLASS
3110

P. A. OS
Other Cust AT
11

INTERNATIONAL INTEREST

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

PROCUREMENT SPECIFICATION
FF-B-187

SUPERSEDES:

MILITARY STANDARD

MS19081

SHEET 1 OF 13

3110-0468

DD FORM 672-1 (COORDINATED)

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selections for all new engineering and design applications and for repetitive use shall be made from this document.

REVIEWER ACTIVITIES: ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES: MC, YD

Dash No.	AFBMAⒷ bearing number (See Note 12)	A	B	C	D	E	R Ⓑ Max. shaft fillet radius	Y Ⓑ Max. housing fillet radius	T	P	S	O	K factor	Basic dynamic load ratings (lb)Ⓑ		Ⓑ Effective load center
	Cone–Cup	Bore	Outside dia.	Bearing width	Cone width	Cup width			ⒷRecommended shoulder diameter Shaft		Housing			Radial	Thrust	
19	1780–1729	1.0000	2.2400	0.7625	0.7810	0.6250	0.03	0.05	1.20	1.18	1.93	2.01	1.90	1,740	915	-0.27
20	15101–15245	1.0000	2.4409	0.7500	0.8125	0.5625	0.03	0.05	1.28	1.24	2.17	2.28	1.67	1,990	1,190	-0.23
21	2687–2631	1.0000	2.6150	0.9375	1.0013	0.7500	0.05	0.05	1.32	1.24	2.28	2.36	2.30	2,970	1,290	-0.37
*22	3189–3120	1.0000	2.8593	1.1875	1.1810	0.9375	0.03	0.13	1.40	1.38	2.40	2.64	1.76	3,700	2,100	-0.40
23	15590–15520	1.1250	2.2500	0.6875	0.6875	0.5313	0.14	0.06	1.56	1.32	2.01	2.09	1.69	1,800	1,070	-0.20
*24	15113–15243	1.1250	2.4375	0.7500	0.8125	0.5625	0.03	0.08	1.36	1.34	2.13	2.28	1.67	1,990	1,190	-0.23
25	2689–2631	1.1250	2.6150	0.9375	1.0013	0.7500	0.05	0.05	1.42	1.34	2.28	2.36	2.30	2,970	1,290	-0.37
26	3198–3120	1.1250	2.8593	1.1875	1.1810	0.9375	0.05	0.13	1.54	1.46	2.40	2.64	1.76	3,700	2,100	-0.40
*27	17119–17244	1.1875	2.4409	0.6300	0.6522	0.5625	0.06	0.06	1.46	1.36	2.13	2.24	1.53	1,620	1,060	-0.14
185	LM67048–LM67010	1.2500	2.3280	0.6250	0.6600	0.4650	***	0.05	1.67	1.42	2.05	2.20	1.42	1,580	1,110	-0.12
28	15126–15245	1.2500	2.4409	0.7500	0.8125	0.5625	0.03	0.05	1.46	1.44	2.17	2.28	1.67	1,990	1,190	-0.23
*29	02476–02421	1.2500	2.6875	0.8750	0.8750	0.6875	0.03	0.03	1.54	1.52	2.32	2.48	1.40	2,490	1,780	-0.20
*186	14125A–14276	1.2500	2.7170	0.7813	0.7710	0.6250	0.14	0.05	1.73	1.48	2.36	2.48	1.53	2,180	1,420	-0.17
30	2580–2523	1.2500	2.7500	0.9375	0.9983	0.7500	0.03	0.05	1.52	1.48	2.40	2.52	2.14	3,280	1,530	-0.34
*31	3188–3130	1.2500	2.8593	1.1875	1.1810	0.9375	0.03	0.03	1.57	1.56	2.48	2.64	1.76	3,700	2,100	-0.40
*187	HM88542–HM88510	1.2500	2.8750	1.1563	1.0938	0.9063	0.05	0.13	1.79	1.68	2.32	2.76	1.07	3,640	3,400	-0.22
32	3476–3420	1.2500	3.1250	1.1563	1.1721	0.9375	0.05	0.13	1.69	1.61	2.64	2.91	1.60	4,150	2,590	-0.34
33	M88048–M88010	1.3125	2.6875	0.8750	0.8750	0.6875	0.03	0.06	1.67	1.62	2.28	2.56	1.07	2,620	2,450	-0.11
*34	HM88547–HM88510	1.3125	2.8750	1.1563	1.0938	0.9063	0.03	0.13	1.79	1.68	2.32	2.76	1.07	3,640	3,400	-0.22
35	HM89443–HM89410	1.3125	3.0000	1.1563	1.1250	0.9063	0.03	0.13	1.83	1.75	2.44	2.87	1.07	3,880	3,630	-0.22
188	LM48548–LM48510	1.3750	2.5625	0.7100	0.7200	0.5500	0.14	0.05	1.81	1.57	2.28	2.40	1.55	2,140	1,380	-0.15
*36	M38549–M38510	1.3750	2.6250	0.8125	0.8125	0.6563	0.14	0.09	1.83	1.57	2.28	2.44	1.66	2,520	1,520	-0.22
37	14137A–14276	1.3750	2.7170	0.7813	0.7710	0.6250	0.06	0.05	1.65	1.57	2.36	2.48	1.53	2,180	1,420	-0.17
*38	2878–2820	1.3750	2.8750	0.8750	0.9375	0.6875	0.03	0.13	1.65	1.61	2.48	2.68	1.59	2,910	1,830	-0.22
39	25877–25821	1.3750	2.8750	0.9375	0.9688	0.7500	0.06	0.03	1.69	1.59	2.56	2.68	2.01	3,340	1,660	-0.32
*40	23690–23620	1.3750	2.8750	1.0625	1.0620	0.8750	0.14	0.06	1.93	1.66	2.48	2.68	1.58	3,540	2,240	-0.32
41	HM89446–HM89410	1.3750	3.0000	1.1563	1.1250	0.9063	0.14	0.13	2.09	1.75	2.44	2.87	1.07	3,880	3,630	-0.22
42	3482–3420	1.3750	3.1250	1.1563	1.1721	0.9375	0.03	0.13	1.73	1.71	2.64	2.91	1.60	4,150	2,590	-0.34

For NOTES see sheets 11 and 12.

FED. SUP CLASS
3110

APPROVED 30 April 1959 REVISED Ⓑ For changes see sheets 1 thru 13

MILITARY STANDARD

MS19081

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

SHEET 2 OF 13

3110-0468

P.A. OS

Other Cust AT
11

INTERNATIONAL INTEREST

SUPERSEDES:

PROCUREMENT SPECIFICATION FF-B-187

DD FORM 672-1 COORDINATED:

This standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

REVIEWER ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES MC, YD

Dash No.	AFBMA Ⓑ bearing number (See Note 12) Cone–Cup	A Bore	B Outside dia.	C Bearing width	D Cone width	E Cup width	R Ⓑ Max. shaft fillet radius	Y Ⓑ Max. housing fillet radius	T Ⓑ Recommended shoulder diameter Shaft	P Shaft	S Housing	O Housing	K factor	Basic dynamic load ratings (lb) Ⓑ Radial	 Thrust	Ⓑ Effective load center
•43	3581–3526	1.3750	3.4375	1.1875	1.2160	0.9375	0.14	0.03	1.95	1.69	2.99	3.15	1.91	4,550	2,380	-0.40
44	HM89449–HM89410	1.4375	3.0000	1.1563	1.1250	0.9063	0.14	0.13	2.13	1.75	2.44	2.87	1.07	3,880	3,630	-0.22
189	31597–31520	1.4375	3.0000	1.1563	1.1250	0.9375	0.14	0.13	2.01	1.75	2.52	2.83	1.45	3,770	2,600	-0.30
190	3878–3820	1.4375	3.3750	1.1875	1.1875	0.9375	0.03	0.13	1.89	1.85	2.87	3.19	1.45	5,000	3,450	-0.32
45	13685–13621	1.5000	2.7170	0.7500	0.7500	0.5938	0.14	0.09	1.95	1.69	2.40	2.56	1.45	2,230	1,540	-0.12
•191	2788A–2735X	1.5000	2.8750	0.9375	1.0100	0.7500	0.06	0.03	1.81	1.71	2.60	2.72	1.93	3,460	1,790	-0.31
46	28150–28300	1.5000	3.0000	0.8125	0.8244	0.6105	0.06	0.05	1.79	1.71	2.68	2.80	1.45	2,440	1,680	-0.19
47	2788–2729	1.5000	3.0000	0.9375	1.0100	0.7500	0.14	0.03	1.97	1.71	2.68	2.76	1.93	3,460	1,790	-0.31
•48	347–332	1.5000	3.1496	0.8268	0.8820	0.7018	0.14	0.05	1.97	1.73	2.87	2.95	2.14	3,000	1,400	-0.25
•49	27881–27820	1.5000	3.1510	0.9688	0.9330	0.7288	0.14	0.06	2.09	1.85	2.68	2.95	1.04	3,080	2,960	-0.10
•50	26878–26820	1.5000	3.1562	1.0000	1.0000	0.8125	0.03	0.13	1.77	1.75	2.72	2.91	1.83	3,660	2,000	-0.29
51	3490–3422	1.5000	3.1562	1.1563	1.1721	0.9375	0.14	0.13	2.05	1.81	2.68	2.91	1.60	4,150	2,590	-0.34
•52	HM801346–HM801310	1.5000	3.2500	1.1563	1.1250	0.9063	0.03	0.13	2.01	1.93	2.68	3.07	1.07	4,300	4,000	-0.19
•53	3583–3525	1.5000	3.4375	1.1875	1.2160	0.9375	0.14	0.13	2.05	1.79	2.95	3.19	1.91	4,550	2,380	-0.40
•54	49150–49368	1.5000	3.6875	1.2500	1.2500	1.0000	0.14	0.13	2.13	1.89	3.23	3.43	1.62	5,250	3,230	-0.36
•55	542–532A	1.5000	4.3750	1.5000	1.4550	1.1875	0.14	0.13	2.17	1.93	3.74	3.94	1.97	6,900	3,500	-0.48
192	2789–2735X	1.5625	2.8750	0.9375	1.0100	0.7500	0.14	0.03	2.05	1.77	2.60	2.72	1.93	3,460	1,790	-0.31
•56	2789–2720	1.5625	3.0000	0.9375	1.0100	0.7500	0.14	0.13	2.05	1.77	2.60	2.76	1.93	3,460	1,790	-0.31
57	3382–3320	1.5625	3.1562	1.1563	1.1965	0.9375	0.14	0.13	2.05	1.79	2.76	2.95	2.14	4,300	2,000	-0.43
.58	18590–18520	1.6250	2.8750	0.6562	0.6875	0.5000	0.14	0.06	2.09	1.81	2.60	2.72	1.67	2,020	1,210	-0.11
59	LM501349–LM501310	1.6250	2.8910	0.7700	0.7800	0.5800	0.14	0.03	2.09	1.83	2.64	2.76	1.46	2,520	1,720	-0.13
60	342–332	1.6250	3.1496	0.8268	0.8820	0.7018	0.14	0.05	2.09	1.81	2.87	2.95	2.14	3,000	1,400	-0.25
61	26882–26820	1.6250	3.1562	1.0000	1.0000	0.8125	0.14	0.13	2.13	1.85	2.72	2.91	1.83	3,660	2,000	-0.29
193	3877–3821	1.6250	3.3750	1.1875	1.1875	0.9375	0.14	0.05	2.24	1.98	2.95	3.19	1.45	5,000	3,450	-0.32
62	3577–3525	1.6250	3.4375	1.1875	1.2160	0.9375	0.14	0.13	2.13	1.89	2.95	3.19	1.91	4,550	2,380	-0.40
•63	49162–49368	1.6250	3.6875	1.2500	1.2500	1.0000	0.14	0.13	2.24	1.97	3.23	3.43	1.62	5,250	3,230	-0.36
64	526–522	1.6250	4.0000	1.3750	1.4200	1.0625	0.14	0.13	2.24	1.97	3.50	3.74	2.05	6,550	3,200	-0.50
65	HM807035–HM807010	1.6250	4.1250	1.4375	1.4375	1.1250	0.06	0.13	2.36	2.24	3.50	3.94	1.20	7,150	5,950	-0.29
66	L102849–L102810	1.7500	2.8750	0.7188	0.7188	0.5938	0.06	0.06	2.01	1.93	2.60	2.72	1.83	2,470	1,350	-0.15
67	12175–12303	1.7500	3.0312	0.6875	0.6750	0.4688	0.06	0.06	2.05	1.95	2.68	2.87	1.15	2,000	1,740	-0.00
•68	35176–35326	1.7500	3.2650	0.8750	0.9060	0.6875	0.03	0.03	1.97	1.95	2.99	3.07	1.96	3,150	1,610	-0.24

For NOTES see sheets 11 and 12.

APPROVED 30 April 1959 REVISED Ⓑ For changes see sheets 1 thru 13

FED. SUP CLASS
3110

P. A. OS
Other Cust. AT
11

INTERNATIONAL INTEREST

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

MILITARY STANDARD
MS19081

SHEET 3 OF 13

3110-0468

PROCUREMENT SPECIFICATION
FF-B-187

SUPERSEDES:

DD FORM 672-1 (COORDINATED)

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

REVIEWER ACTIVITIES: ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES: MC, YD

Dash No.	AFBMAⒷ bearing number (See Note 12)	A	B	C	D	E	R	Y	T	P	S	O	K factor	Basic dynamic load ratings (lb)Ⓑ		Ⓑ Effective load center
	Cone–Cup	Bore	Outside dia.	Bearing width	Cone width	Cup width	Ⓑ Max. shaft fillet radius	Ⓑ Max. housing fillet radius	ⒷRecommended shoulder diameter Shaft		Housing			Radial	Thrust	
69	25580–25520	1.7500	3.2650	0.9375	1.0000	0.7500	0.14	0.03	2.24	1.97	2.91	3.03	1.74	3,680	2,110	-0.25
194	25580–25521	1.7500	3.2700	0.9375	1.0000	0.7500	0.14	0.13	2.24	1.97	2.83	3.03	1.74	3,680	2,110	-0.25
70	3578–3525	1.7500	3.4375	1.1875	1.2160	0.9375	0.14	0.13	2.24	2.01	2.95	3.19	1.91	4,550	2,380	-0.40
•71	49175–49368	1.7500	3.6875	1.2500	1.2500	1.0000	0.14	0.13	2.32	2.09	3.23	3.43	1.62	5,250	3,230	-0.36
•195	HM804843–HM804810	1.7500	3.7500	1.1875	1.1563	0.9063	0.14	0.13	2.48	2.24	3.19	3.58	1.07	5,150	4,800	-0.15
•196	3783–3732	1.7500	3.8750	1.1875	1.1930	0.9375	0.25	0.13	2.52	2.05	3.31	3.54	1.73	5,000	2,890	-0.32
72	527–522	1.7500	4.0000	1.3750	1.4200	1.0625	0.14	0.13	2.32	2.09	3.50	3.74	2.05	6,550	3,200	-0.50
73	HM807040–HM807010	1.7500	4.1250	1.4375	1.4375	1.1250	0.14	0.13	2.60	2.32	3.50	3.94	1.20	7,150	5,950	-0.29
•74	535–532X	1.7500	4.2500	1.4375	1.4550	1.1250	0.14	0.13	2.36	2.13	3.70	3.94	1.97	6,900	3,500	-0.48
•75	615–612A	1.7500	4.7260	1.6250	1.6250	1.2500	0.14	0.13	2.44	2.20	4.06	4.33	1.86	8,300	4,450	-0.55
76	LM603049–LM603011	1.7812	3.0625	0.7812	0.7812	0.5937	0.14	0.03	2.24	1.97	2.80	2.91	1.37	2,610	1,910	-0.09
77	25590–25523	1.7960	3.2650	1.0625	1.0000	0.8750	0.14	0.09	2.28	2.01	2.83	3.03	1.74	3,680	2,110	-0.25
•197	18690–18620	1.8125	3.1250	0.6875	0.6875	0.5313	0.11	0.06	2.20	2.01	2.80	2.91	1.56	2,090	1,340	-0.08
78	2984–2924	1.8125	3.3465	1.0000	1.0082	0.8125	0.14	0.05	2.28	2.05	2.99	3.15	1.69	3,830	2,270	-0.25
79	3779–3720	1.8750	3.6718	1.1875	1.1930	0.9375	0.14	0.13	2.40	2.17	3.23	3.46	1.73	5,000	2,890	-0.32
198	HM804846–HM804010	1.8750	3.7500	1.1875	1.1563	0.9063	0.14	0.13	2.60	2.27	3.19	3.58	1.07	5,150	4,800	-0.15
199	528–522	1.8750	4.0000	1.3750	1.4200	1.0625	0.14	0.13	2.44	2.17	3.50	3.74	2.05	6,550	3,200	-0.50
•80	617–612A	1.8750	4.7260	1.6250	1.6250	1.2500	0.14	0.13	2.56	2.28	4.06	4.33	1.86	8,300	4,450	-0.55
200	HM804849–HM804810	1.9060	3.7500	1.1875	1.1563	0.9063	0.14	0.13	2.60	2.27	3.19	3.58	1.07	5,150	4,800	-0.15
201	5395–5335	1.9375	4.0625	1.7188	1.7510	1.4375	0.14	0.13	2.60	2.36	3.50	3.82	1.97	8,800	4,450	-0.63
202	HM807044–HM807010	1.9375	4.1250	1.4375	1.4375	1.1250	0.14	0.13	2.72	2.48	3.50	3.94	1.20	7,150	5,950	-0.29
•81	65390–65320	1.9375	4.5000	1.7500	1.7500	1.3750	0.14	0.13	2.76	2.36	3.82	4.21	1.36	8,850	6,500	-0.49
203	HH506348–HH506310	1.9375	4.5000	1.7500	1.7500	1.4200	0.14	0.13	2.80	2.40	3.82	4.21	1.45	9,900	6,800	-0.53
82	LL205449–LL205410	2.0000	3.0625	0.5000	0.5000	0.3750	0.06	0.06	2.24	2.17	2.80	2.91	1.74	1,500	860	0.00
83	L305649–L305610	2.0000	3.1875	0.7188	0.7188	0.5625	0.06	0.06	2.28	2.20	2.87	3.03	1.64	2,680	1,630	-0.09
84	18790–18720	2.0000	3.3465	0.6875	0.6875	0.5313	0.14	0.06	2.44	2.20	3.03	3.15	1.44	2,240	1,550	-0.03
85	368–362	2.0000	3.5433	0.7874	0.8750	0.6250	0.06	0.08	2.28	2.20	3.19	3.31	1.83	3,310	1,810	-0.17
204	3780–3730	2.0000	3.6718	1.1875	1.1930	0.9375	0.14	0.03	2.52	2.28	3.31	3.46	1.73	5,000	2,890	-0.32
86	3780–3720	2.0000	3.6718	1.1875	1.1930	0.9375	0.14	0.13	2.52	2.28	3.23	3.46	1.73	5,000	2,890	-0.32
205	49585–49522	2.0000	4.0000	1.2500	1.2500	1.0000	0.14	0.03	2.60	2.32	3.54	3.78	1.46	5,300	3,630	-0.28

For NOTES see sheets 11 and 12.

APPROVED 30 April 1959 REVISED Ⓑ For changes see sheets 1 thru 13

FED. SUP CLASS
3110

P. A. OS
Other Cust AT
11

INTERNATIONAL INTEREST

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

SUPERSEDES:

PROCUREMENT SPECIFICATION
FF-B-187

MILITARY STANDARD

MS19081

SHEET 4 OF 13

3110-0468

DD FORM 672-1

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

REVIEW ACTIVITIES ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES: MC, YD

Dash No.	AFBMAⒷ bearing number (See Note 12) Cone–Cup	A Bore	B Outside dia.	C Bearing width	D Cone width	E Cup width	R Ⓑ Max. shaft fillet radius	Y Ⓑ Max. housing fillet radius	T ⒷRecommended shoulder diameter Shaft	P Shaft	S Housing	O Housing	K factor	Basic dynamic load ratings (lb)Ⓑ Radial	Thrust	Ⓑ Effective load center
206	49585–49520	2.0000	4.0000	1.2500	1.2500	1.0000	0.14	0.13	2.60	2.32	3.46	3.78	1.46	5,300	3,630	-0.28
207	529–522	2.0000	4.0000	1.3750	1.4200	1.0625	0.03	0.13	2.32	2.28	3.50	3.74	2.05	6,550	3,200	-0.50
87	529X–522	2.0000	4.0000	1.3750	1.4200	1.0625	0.14	0.13	2.56	2.28	3.50	3.74	2.05	6,550	3,200	-0.50
208	45284–45220	2.0000	4.1250	1.1875	1.2188	0.9375	0.25	0.13	2.80	2.32	3.66	3.90	1.76	6,200	3,520	-0.32
88	HM807046–HM807010	2.0000	4.1250	1.4375	1.4375	1.1250	0.14	0.13	2.76	2.48	3.50	3.94	1.20	7,150	5,950	-0.29
209	537–532X	2.0000	4.2500	1.4375	1.4550	1.1250	0.14	0.13	2.56	2.32	3.70	3.94	1.97	6,900	3,500	-0.48
•89	619–612A	2.0000	4.7260	1.6250	1.6250	1.2500	0.14	0.13	2.64	2.40	4.06	4.33	1.86	8,300	4,450	-0.55
210	555-553X	2.0000	4.8125	1.5000	1.4440	1.1875	0.09	0.13	2.60	2.44	4.25	4.53	1.69	7,900	4,700	-0.37
90	65200–65500	2.0000	5.0000	1.7500	1.7500	1.3750	0.14	0.13	2.95	2.72	4.21	4.69	1.20	9,850	8,200	-0.37
211	33895–33821	2.1250	3.7500	1.0938	1.1250	0.8750	0.06	0.09	2.48	2.36	3.35	3.54	1.77	5,250	2,960	-0.30
212	456–453X	2.1250	4.1250	1.1875	1.1542	0.9687	0.14	0.13	2.68	2.40	3.62	3.86	1.74	5,400	3,090	-0.28
91	HM807049–HM807010	2.1250	4.1250	1.4375	1.4375	1.1250	0.14	0.13	2.87	2.48	3.50	3.94	1.20	7,150	5,950	-0.29
92	539–532A	2.1250	4.3750	1.5000	1.4550	1.1875	0.14	0.13	2.68	2.40	3.74	3.94	1.97	6,900	3,500	-0.48
213	621–612	2.1250	4.7500	1.6250	1.6250	1.2500	0.14	0.13	2.76	2.48	4.13	4.33	1.86	8,300	4,450	-0.55
214	65212–65500	2.1250	5.0000	1.7500	1.7500	1.3750	0.14	0.13	3.03	2.80	4.21	4.69	1.20	9,850	8,200	-0.37
215	6280–6220	2.1250	5.0000	2.0000	2.0625	1.6250	0.14	0.13	2.91	2.64	4.25	4.61	1.96	12,400	6,350	-0.77
216	5566–5535	2.1875	4.8125	1.7188	1.7230	1.4375	0.05	0.13	2.76	2.68	4.17	4.57	1.63	10,200	6,250	-0.48
•217	HM813840–HM813810	2.1875	5.0000	1.4375	1.4375	1.0625	0.14	0.13	2.99	2.76	4.37	4.76	1.16	8,050	6,950	-0.15
93	L507949–L507910	2.2500	3.4375	0.7188	0.7188	0.5625	0.06	0.06	2.56	2.44	3.11	3.27	1.50	2,810	1,870	-0.03
241	387A–382A	2.2500	3.8125	0.8268	0.8640	0.6250	0.14	0.03	2.72	2.44	3.50	3.62	1.65	3,590	2,180	-0.12
179	28682–28622	2.2500	3.8437	0.9688	0.9688	0.7656	0.14	0.03	2.76	2.48	3.46	3.62	1.45	4,400	3,040	-0.13
94	387–382	2.2500	3.8750	0.8268	0.8640	0.7018	0.09	0.03	2.60	2.44	3.54	3.62	1.65	3,590	2,180	-0.12
218	462–453X	2.2500	4.1250	1.1875	1.1542	0.9687	0.09	0.13	2.64	2.48	3.62	3.86	1.74	5,400	3,090	-0.28
95	45290–45220	2.2500	4.1250	1.1875	1.2188	0.9375	0.09	0.13	2.68	2.56	3.66	3.90	1.76	6,200	3,520	-0.32
96	390–394A	2.2500	4.3307	0.8661	0.8660	0.7411	0.09	0.05	2.76	2.60	3.98	4.11	1.45	4,000	2,770	-0.03
•97	462–454	2.2500	4.3307	1.0943	1.1542	1.0630	0.09	0.08	2.64	2.48	3.78	3.94	1.74	5,400	3,090	-0.28
242	3979–3920	2.2500	4.4375	1.1875	1.1830	0.9375	0.14	0.13	2.83	2.60	3.90	4.17	1.45	5,900	4,050	-0.18
243	33225–33462	2.2500	4.6250	1.1875	1.1875	0.9375	0.14	0.13	2.91	2.68	4.09	4.41	1.34	5,950	4,450	-0.11
98	623–612	2.2500	4.7500	1.6250	1.6250	1.2500	0.14	0.13	2.83	2.60	4.13	4.33	1.86	8,300	4,450	-0.55
99	65225–65500	2.2500	5.0000	1.7500	1.7500	1.3750	0.14	0.13	3.15	2.80	4.21	4.69	1.20	9,850	8,200	-0.37

For NOTES see sheets 11 and 12.

APPROVED 30 April 1959 REVISED Ⓑ For changes see sheets 1 thru 12

FED. SUP CLASS
3110

P. A. OS
Other Cust AT
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INTERNATIONAL INTEREST

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

MILITARY STANDARD

MS19081

PROCUREMENT SPECIFICATION
FF-B-187

SUPERSEDES:

SHEET 5 OF 13

3110-0468

DD FORM 672-1 •CONTINUATION•

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REVIEWER ACTIVITIES ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES MC, YD

Dash No.	AFBMAⒷ bearing number (See Note 12)	A	B	C	D	E	R Ⓑ	Y Ⓑ	T	P	S	O	K factor	Basic dynamic load ratings (lb)Ⓑ		aⒷ
	Cone–Cup	Bore	Outside dia.	Bearing width	Cone width	Cup width	Max. shaft fillet radius	Max. housing fillet radius	Ⓑ Recommended shoulder diameter Shaft		Housing			Radial	Thrust	Effective load center
•100	6375–6320	2.2500	5.3447	2.1250	2.2050	1.7500	0.17	0.13	3.15	2.83	4.61	4.96	1.80	13,300	7,350	-0.76
219	6455–6420	2.2500	5.8750	2.1250	2.1350	1.7500	0.14	0.13	3.19	2.95	5.08	5.51	1.61	14,700	9,100	-0.59
220	3981–3920	2.3125	4.4375	1.1875	1.1830	0.9375	0.14	0.13	2.87	2.64	3.90	4.17	1.45	5,900	4,050	-0.18
101	28985–28920	2.3750	4.0000	1.0000	1.0000	0.7813	0.14	0.13	2.87	2.64	3.54	3.82	1.37	4,550	3,310	-0.10
221	HM212044–HM212010	2.3750	4.8125	1.5000	1.5100	1.1700	0.31	0.06	3.35	2.76	4.33	4.57	1.73	9,150	5,300	-0.43
102	65237–65500	2.3750	5.0000	1.7500	1.7500	1.3750	0.14	0.13	3.23	2.80	4.21	4.69	1.20	9,850	8,200	-0.37
222	637–633	2.3750	5.1250	1.6250	1.6250	1.2500	0.14	0.13	3.07	2.83	4.57	4.88	1.61	9,600	5,950	-0.44
223	L610549–L610510	2.5000	3.1788	0.7500	0.7500	0.5938	0.06	0.06	2.80	2.72	3.39	3.58	1.38	3,070	2,220	+0.02
224	29585–29520	2.5000	4.2500	1.0000	1.0000	0.7500	0.14	0.13	3.03	2.80	3.78	4.06	1.27	4,650	3,670	-0.03
103	390A–394A	2.5000	4.3307	0.8661	0.8660	0.7411	0.06	0.05	2.87	2.76	3.98	4.11	1.45	4,000	2,770	-0.03
•104	29586–29521	2.5000	4.3307	1.0000	1.0000	0.7500	0.06	0.05	2.87	2.80	3.90	4.09	1.27	4,650	3,670	-0.03
105	39585–39520	2.5000	4.4375	1.1875	1.1875	0.9375	0.14	0.13	3.03	2.80	3.98	4.21	1.72	7,000	4,050	-0.26
106	5584–5535	2.5000	4.8125	1.7188	1.7230	1.4375	0.14	0.13	3.19	2.95	4.17	4.57	1.63	10,200	6,250	-0.48
107	565–563	2.5000	5.0000	1.4375	1.4240	1.1250	0.14	0.13	3.15	2.87	4.41	4.72	1.61	8,250	5,100	-0.32
225	HM813842–HM813811	2.5000	5.0000	1.4375	1.4375	1.0625	0.14	0.06	3.23	2.99	4.45	4.76	1.16	8,050	6,950	-0.15
•108	639–633	2.5000	5.1250	1.6250	1.6250	1.2500	0.14	0.13	3.19	2.91	4.57	4.88	1.61	9,600	5,950	-0.44
109	H414235–H414210	2.5000	5.3750	1.6250	1.6250	1.2500	0.14	0.13	3.23	3.07	4.76	5.08	1.62	10,900	6,750	-0.43
110	H715336–H715311	2.5000	5,3750	1.8125	1.8125	1.4375	0.14	0.13	3.43	3.15	4.65	5.20	1.24	11,900	9,600	-0.33
•111	6475–6420	2.5000	5.8750	2.1250	2.1350	1.7500	0.14	0.13	3.39	3.15	5.08	5.51	1.61	14,700	9,100	-0.59
226	569–563	2.5576	5.0000	1.4375	1.4240	1.1250	0.14	0.13	3.19	2.91	4.41	4.72	1.61	8,250	5,100	-0.32
227	6379–6320	2.5625	5.3447	2.1250	2.2050	1.7500	0.14	0.13	3.31	3.05	4.61	4.96	1.80	13,300	7,350	-0.76
228	395S–394A	2.6250	4.3307	0.8661	0.8660	0.7411	0.14	0.05	3.11	2.87	3.98	4.11	1.45	4,000	2,770	-0.03
112	3984–3920	2.6250	4.4375	1.1875	1.1830	0.9375	0.14	0.13	3.15	2.91	3.90	4.17	1.45	5,900	4,050	-0.18
•229	3994–3920	2.6250	4.4375	1.1875	1.1830	0.9375	0.22	0.13	3.31	2.91	3.90	4.17	1.45	5,900	4,050	-0.18
230	39590–39520	2.6250	4.4375	1.1875	1.1875	0.9375	0.14	0.13	3.15	2.91	3.98	4.21	1.72	7,000	4,050	-0.26
231	39591–39520	2.6250	4.4375	1.1875	1.1875	0.9375	0.22	0.13	3.31	2.91	3.98	4.21	1.72	7,000	4,050	-0.26
232	33262–33462	2.6250	4.6250	1.1875	1.1875	0.9375	0.14	0.13	3.19	2.95	4.09	4.41	1.34	5,950	4,450	-0.11
233	560–553X	2.6250	4.8125	1.5000	1.4440	1.1875	0.14	0.13	3.19	2.95	4.25	4.53	1.69	7,900	4,700	-0.37
234	HM212049–HM212010	2.6250	4.8125	1.5000	1.5100	1.1700	0.14	0.06	3.23	2.97	4.33	4.57	1.73	9,150	5,300	-0.43
113	HM212049–HM212011	2.6250	4.8125	1.5000	1.5100	1.1700	0.14	0.13	3.23	2.97	4.25	4.57	1.73	9,150	5,300	-0.43

For NOTES see sheets 11 and 12.

FED. SUP CLASS
3110

APPROVED 30 April 1959 REVISED Ⓑ For changes see sheets 1 thru 13

P. A. OS
Other Cust AT
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INTERNATIONAL INTEREST

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

SUPERSEDES:

PROCUREMENT SPECIFICATION
FF-B-187

MILITARY STANDARD

MS19081

SHEET 6 OF 13

3110-0468

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REVIEWER ACTIVITIES: ME, MI, AV, MU, IS, NS, SH USER ACTIVITIES: MC, YD

Dash No.	AFBMAⒷ bearing number (See Note 12)	A	B	C	D	E	R Ⓑ	Y Ⓑ	T	P	S	O	K factor	Basic dynamic load ratings (lb)Ⓑ		aⒷ Effective load center
	Cone–Cup	Bore	Outside dia.	Bearing width	Cone width	Cup width	Max. shaft fillet radius	Max. housing fillet radius	ⒷRecommended shoulder diameter					Radial	Thrust	
									Shaft		Housing					
235	H414242–H414210	2.6250	5.3750	1.6250	1.6250	1.2500	0.14	0.13	3.35	3.19	4.76	5.08	1.62	10,900	6,750	-0.43
236	H715341–H715311	2.6250	5.3750	1.8125	1.8125	1.4375	0.14	0.13	3.50	3.27	4.65	5.20	1.24	11,900	9,600	-0.33
114	399A–394A	2.6875	4.3307	0.8661	0.8660	0.7411	0.09	0.05	3.07	2.91	3.98	4.11	1.45	4,000	2,770	-0.03
237	399AS–394A	2.6875	4.3307	0.8661	0.8660	0.7411	0.20	0.05	3.27	2.91	3.98	4.11	1.45	4,000	2,770	-0.03
238	33269–33462	2.6875	4.6250	1.1875	1.1875	0.9375	0.14	0.13	3.23	2.99	4.09	4.41	1.34	5,950	4,450	-0.11
239	480–472A	2.6875	4.7244	1.1418	1.1420	0.9230	0.14	0.13	3.23	2.95	4.17	4.49	1.52	5,900	3,900	-0.16
240	560S–553X	2.6875	4.8125	1.5000	1.4440	1.1875	0.14	0.13	3.27	2.99	4.25	4.53	1.69	7,900	4,700	-0.37
250	H715343–H715311	2.6875	5.3750	1.8125	1.8125	1.4375	0.14	0.13	3.54	3.31	4.65	5.20	1.24	11,900	9,600	-0.33
251	29675–29620	2.7500	4.4375	1.0000	1.0000	0.7500	0.06	0.13	3.15	3.03	3.98	4.29	1.20	4,900	4,100	0.04
252	33275–33462	2.7500	4.6250	1.1875	1.1875	0.9375	0.14	0.13	3.31	3.03	4.09	4.41	1.34	5,950	4,450	-0.11
115	47487–47420	2.7500	4.7244	1.2813	1.2813	1.0313	0.14	0.13	3.31	3.07	4.21	4.49	1.62	7,600	4,700	-0.25
116	566–563	2.7500	5.0000	1.4375	1.4240	1.1250	0.14	0.13	3.35	3.07	4.41	4.72	1.61	8,250	5,100	-0.32
253	HM813846–HM813811	2.7500	5.0000	1.4375	1.4375	1.0625	0.14	0.06	3.46	3.19	4.45	4.76	1.16	8,050	6,950	-0.15
180	643–633	2.7500	5.1250	1.6250	1.6250	1.2500	0.14	0.13	3.39	3.15	4.57	4.88	1.61	9,600	5,950	-0.44
117	643–632	2.7500	5.3750	1.6250	1.6250	1.2500	0.14	0.13	3.39	3.15	4.65	4.92	1.61	9,600	5,950	-0.44
254	655–653	2.7500	5.7500	1.6250	1.6250	1.2500	0.14	0.13	3.46	3.23	5.16	5.47	1.43	10,300	7,150	-0.31
•118	6454–6420	2.7500	5.8750	2.1250	2.1350	1.7500	0.20	0.13	3.70	3.35	5.08	5.51	1.61	14,700	9,100	-0.59
255	745A–742	2.7500	5.9090	1.7500	1.8375	1.4375	0.14	0.13	3.46	3.23	5.28	5.59	1.80	13,200	7,350	-0.47
256	34274–34478	2.7340	4.7812	0.9688	0.9060	0.6875	0.08	0.08	3.19	3.07	4.33	4.57	1.30	4,250	3,270	0.06
257	33281–33462	2.8125	4.6250	1.1875	1.1875	0.9375	0.14	0.13	3.35	3.11	4.09	4.41	1.34	5,950	4,450	-0.11
258	47490–47420	2.8125	4.7244	1.2813	1.2813	1.0313	0.14	0.13	3.39	3.11	4.21	4.49	1.62	7,600	4,700	-0.25
119	H414249–H414210	2.8125	5.3750	1.6250	1.6250	1.2500	0.14	0.13	3.50	3.28	4.76	5.08	1.62	10,900	6,750	-0.43
259	H715345–H715311	2.8125	5.3750	1.8125	1.8125	1.4375	0.14	0.13	3.66	3.43	4.65	5.20	1.24	11,900	9,600	-0.33
•120	H715345–H715310	2.8125	5.5000	1.8125	1.8125	1.4375	0.14	0.13	3.66	3.43	4.72	5.24	1.24	11,900	9,600	-0.33
121	29685–29620	2.8750	4.4375	1.0000	1.0000	0.7500	0.14	0.13	3.39	3.15	3.98	4.29	1.20	4,900	4,100	0.04
122	33287–33462	2.8750	4.6250	1.1875	1.1875	0.9375	0.14	0.13	3.43	3.15	4.09	4.41	1.34	5,950	4,450	-0.11
260	27680–27620	2.8750	4.9375	1.0000	1.0000	0.7813	0.14	0.06	3.46	3.23	4.53	4.72	1.40	5,150	3,690	0.02
123	567–563	2.8750	5.0000	1.4375	1.4240	1.1250	0.14	0.13	3.46	3.19	4.41	4.72	1.61	8,250	5,100	-0.32
261	6460–6420	2.8750	5.8750	2.1250	2.1350	1.7500	0.14	0.13	3.66	3.43	5.08	5.51	1.61	14,700	9,100	-0.59
262	744–742	2.8750	5.9090	1.7500	1.8375	1.4375	0.14	0.13	3.58	3.35	5.28	5.59	1.80	13,200	7,350	-0.47
263	577–572	2.9375	5.5115	1.4375	1.4212	1.1250	0.14	0.13	3.58	3.35	4.92	5.24	1.45	8,800	6,100	-0.21

For NOTES see sheets 11 and 12.

APPROVED 30 April 1959 REVISEDⒷ For changes see sheets 1 thru 13

FED. SUP CLASS
3110

MILITARY STANDARD

MS19081

SHEET 7 OF 13

3110-0468

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

P. A. OS
Other Cust AT
11

INTERNATIONAL INTEREST

PROCUREMENT SPECIFICATION
FF-B-187

SUPERSEDES:

DD FORM 672-1 (COORDINATED)

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document

REVIEWER ACTIVITIES: ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES: MC, YD

Dash No.	AFBMA Ⓑ bearing number (See Note 12)	A	B	C	D	E	R	Y	T	P	S	O	K factor	Basic dynamic load ratings (lb) Ⓑ		Ⓑ Effective load center
	Cone–Cup	Bore	Outside dia.	Bearing width	Cone width	Cup width	Ⓑ Max. shaft fillet radius	Ⓑ Max. housing fillet radius	Ⓑ Recommended shoulder diameter					Radial	Thrust	
									Shaft		Housing					
264	27684–27620	3.0000	4.9375	1.0000	1.0000	0.7813	0.14	0.06	3.58	3.31	4.53	4.72	1.40	5,150	3,690	0.02
265	5760–5735	3.0000	5.3438	1.7500	1.8150	1.3750	0.14	0.13	3.70	3.46	4.69	5.12	1.44	11,200	7,800	-0.46
124	575–572	3.0000	5.5115	1.4375	1.4212	1.1250	0.14	0.13	3.62	3.39	4.92	5.24	1.45	8,800	6,100	-0.21
125	659–653	3.0000	5.7500	1.6250	1.6250	1.2500	0.14	0.13	3.66	3.43	5.16	5.47	1.43	10,300	7,150	-0.31
126	6461–6420	3.0000	5.8750	2.1250	2.1350	1.7500	0.14	0.13	3.78	3.52	5.08	5.51	1.61	14,700	9,100	-0.59
266	659–652	3.0000	6.0000	1.6250	1.6250	1.2500	0.14	0.13	3.66	3.43	5.28	5.55	1.43	10,300	7,150	-0.31
267	755–752	3.0000	6.3750	1.8750	1.9000	1.5000	0.14	0.13	3.74	3.46	5.67	5.91	1.71	13,800	8,050	-0.47
127	6576–6535	3.0000	6.3750	2.1250	2.1693	1.6875	0.14	0.13	3.90	3.62	5.55	6.06	1.46	16,000	11,000	-0.52
268	843–832	3.0000	6.6250	2.1250	2.2190	1.6250	0.25	0.13	3.98	3.50	5.87	6.10	1.95	16,900	8,700	-0.73
269	HH221430–HH221410	3.0000	7.5000	2.2500	2.2650	1.8125	0.14	0.13	3.98	3.74	6.73	7.05	1.74	22,000	12,600	-0.59
128	42690–42620	3.0625	5.0000	1.1875	1.2205	0.8750	0.14	0.13	3.58	3.35	4.49	4.76	1.39	6,750	4,850	-0.11
270	5795–5735	3.0625	5.3438	1.7500	1.8150	1.3750	0.14	0.13	3.78	3.50	4.69	5.12	1.44	11,200	7,800	-0.46
271	750–742	3.1250	5.9090	1.7500	1.8375	1.4375	0.14	0.13	3.78	3.54	5.28	5.59	1.80	13,200	7,350	-0.47
272	740–742	3.1875	5.9090	1.7500	1.8375	1.4375	0.20	0.13	3.98	3.58	5.28	5.59	1.80	13,200	7,350	-0.47
244	L116149–L116110	3.2500	4.5625	0.8125	0.8438	0.6563	0.06	0.06	3.54	3.46	4.25	4.37	1.90	4,150	2,170	-0.05
245	27687–27620	3.2500	4.9375	1.0000	1.0000	0.7813	0.14	0.06	3.78	3.50	4.53	4.72	1.40	5,150	3,690	0.02
246	495–492A	3.2500	5.2500	1.1875	1.1720	0.8750	0.14	0.13	3.82	3.54	4.72	5.04	1.31	6,550	5,000	-0.03
129	47686–47620	3.2500	5.2500	1.3125	1.3125	1.0313	0.14	0.13	3.82	3.54	4.69	5.04	1.44	7,850	5,450	-0.17
130	HM516449–HM516410	3.2500	5.2500	1.5625	1.5625	1.2813	0.14	0.13	3.90	3.62	4.65	5.04	1.45	10,000	6,900	-0.29
131	580–572	3.2500	5.5115	1.4375	1.4212	1.1250	0.14	0.13	3.86	3.58	4.92	5.24	1.45	8,800	6,100	-0.21
132	663–653	3.2500	5.7500	1.6250	1.6250	1.2500	0.14	0.13	3.90	3.62	5.16	5.47	1.43	10,300	7,150	-0.31
247	757–752	3.2500	6.3750	1.8750	1.9000	1.5000	0.14	0.13	3.94	3.70	5.67	5.91	1.71	13,800	8,050	-0.47
133	6559–6535	3.2500	6.3750	2.1250	2.1693	1.6875	0.14	0.13	4.09	3.86	5.55	6.06	1.46	16,000	11,000	-0.52
248	842–832	3.2500	6.6250	2.1250	2.2190	1.6250	0.14	0.13	3.98	3.70	5.87	6.10	1.95	16,900	8,700	-0.73
273	27690–27620	3.2813	4.9375	1.0000	1.0000	0.7813	0.14	0.06	3.78	3.54	4.53	4.72	1.40	5,150	3,690	0.02
134	749–742	3.3475	5.9090	1.7500	1.8375	1.4375	0.14	0.13	3.98	3.74	5.28	5.59	1.80	13,200	7,350	-0.47
274	497–492A	3.3750	5.2500	1.1875	1.1720	0.8750	0.14	0.13	3.90	3.66	4.72	5.04	1.31	6,550	5,000	-0.03
135	497–493	3.3750	5.3750	1.1875	1.1720	0.8750	0.14	0.13	3.90	3.66	4.80	5.12	1.31	6,550	5,000	-0.03
•136	HM617049–HM617010	3.3750	5.5960	1.6875	1.6875	1.3438	0.19	0.13	4.17	3.77	4.92	5.39	1.35	11,600	8,600	-0.29
181	596–592A	3.3750	6.0000	1.5625	1.4300	1.1875	0.14	0.13	4.02	3.78	5.31	5.67	1.32	9,350	7,100	-0.10

For NOTES see sheets 11 and 12.

FED. SUP CLASS 3110

APPROVED 30 April 1959 REVISED Ⓑ For changes see sheets 1 thru 13

MILITARY STANDARD

MS19081

SHEET 8 OF 13

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

P.A. OS
Other Cust AT
11

INTERNATIONAL INTEREST

SUPERSEDES:

PROCUREMENT SPECIFICATION
FF-B-187

DD FORM 672-1

3110-0468

…ary standard is approved for use by all Departments and Agencies of the Department of Defense
Selection for all new engineering and design applications and for repetitive use shall be made from this document

REVIE… …ITIES ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES: MC, YD

Dash No.	AFBMA Ⓑ bearing number (See Note 12)	A	B	C	D	E	R	Y	T	P	S	O	K factor	Basic dynamic load ratings (lb) Ⓑ		*Ⓑ Effective load center
	Cone–Cup	Bore	Outside dia.	Bearing width	Cone width	Cup width	Ⓑ Max. shaft fillet radius	Ⓑ Max. housing fillet radius	Ⓑ Recommended shoulder diameter Shaft		Housing			Radial	Thrust	
275	758–752	3.3750	6.3750	1.8750	1.9000	1.5000	0.14	0.13	4.06	3.82	5.67	5.91	1.71	13,800	8,050	-0.47
276	677–672	3.3750	6.6250	1.6250	1.6250	1.1875	0.14	0.13	4.13	3.90	5.87	6.30	1.24	11,500	9,250	-0.11
277	841–832	3.3750	6.6250	2.1250	2.2190	1.6250	0.14	0.13	4.09	3.82	5.87	6.10	1.95	16,900	8,700	-0.73
278	LL217849–LL217810	3.5000	4.7812	0.5938	0.5938	0.4375	0.06	0.06	3.82	3.70	4.53	4.61	1.77	2,600	1,470	0.12
279	42350–42587	3.5000	5.8750	1.2500	1.1406	0.9688	0.12	0.13	4.09	3.86	5.28	5.63	1.19	7,100	6,000	0.12
137	593–592A	3.5000	6.0000	1.5625	1.4300	1.1875	0.14	0.13	4.09	3.86	5.31	5.67	1.32	9,350	7,100	-0.10
182	593A–592A	3.5000	6.0000	1.5625	1.4300	1.1875	0.25	0.13	4.33	3.86	5.31	5.67	1.32	9,350	7,100	-0.10
138	759–752	3.5000	6.3750	1.8750	1.9000	1.5000	0.14	0.13	4.17	3.90	5.67	5.91	1.71	13,800	8,050	-0.47
139	6580–6535	3.5000	6.3750	2.1250	2.1693	1.6875	0.14	0.13	4.29	4.01	5.55	6.06	1.46	16,000	11,000	-0.52
280	679–672	3.5000	6.6250	1.6250	1.6250	1.1875	0.14	0.13	4.21	3.98	5.87	6.30	1.24	11,500	9,250	-0.11
281	850–832	3.5000	6.6250	2.1250	2.2190	1.6250	0.14	0.13	4.17	3.94	5.87	6.10	1.95	16,900	8,700	-0.73
282	77350–77675	3.5000	6.7500	1.8750	1.9000	1.5000	0.20	0.13	4.33	3.98	6.02	6.34	1.59	14,300	9,000	-0.38
140	855–854	3.5000	7.5000	2.2500	2.2650	1.7500	0.31	0.13	4.65	4.06	6.69	6.85	1.74	19,300	11,100	-0.60
283	HM218248–HM218210	3.5433	5.7874	1.5748	1.5748	1.2795	0.28	0.14	4.41	3.90	5.24	5.55	1.76	11,700	6,650	-0.34
141	760–752	3.5625	6.3750	1.8750	1.9000	1.5000	0.14	0.13	4.21	3.98	5.67	5.91	1.71	13,800	8,050	-0.47
284	47890–47820	3.6250	5.7500	1.3125	1.3750	1.0313	0.14	0.13	4.21	3.98	5.16	5.51	1.30	8,750	6,750	-0.04
285	42368–42584	3.6875	5.8437	1.1250	1.1406	0.8438	0.12	0.12	4.21	4.02	5.28	5.59	1.19	7,100	6,000	0.12
286	L319249–319210	3.7500	5.1250	0.8125	0.8438	0.6563	0.06	0.06	4.06	3.98	4.80	4.92	1.67	4,450	2,670	0.05
287	47896–47820	3.7500	5.7500	1.3125	1.3750	1.0313	0.14	0.13	4.33	4.06	5.16	5.51	1.30	8,750	6,750	-0.04
142	594–592A	3.7500	6.0000	1.5625	1.4300	1.1875	0.14	0.13	4.33	4.09	5.31	5.67	1.32	9,350	7,100	-0.10
288	77375–77675	3.7500	6.7500	1.8750	1.9000	1.5000	0.14	0.13	4.45	4.17	6.02	6.34	1.59	14,300	9,000	-0.38
143	776–772	3.7500	7.1250	1.8750	1.8900	1.5000	0.14	0.13	4.49	4.21	6.34	6.61	1.51	14,900	9,850	-0.32
144	864–854	3.7500	7.5000	2.2500	2.2650	1.7500	0.31	0.13	4.84	4.25	6.69	6.85	1.74	19,300	11,100	-0.60
145	HH221440–HH221416	3.7500	7.8750	2.4375	2.2650	2.0000	0.31	0.13	4.92	4.33	6.87	7.05	1.74	22,000	12,600	-0.59
289	685–672	3.8750	6.6250	1.6250	1.6250	1.1875	0.14	0.13	4.57	4.29	5.87	6.30	1.24	11,500	9,250	-0.11
290	779–772	3.8750	7.1250	1.8750	1.8900	1.5000	0.14	0.13	4.57	4.33	6.34	6.61	1.51	14,900	9,850	-0.32
291	866–854	3.8750	7.5000	2.2500	2.2650	1.7500	0.14	0.13	4.65	4.37	6.69	6.85	1.74	19,300	11,100	-0.60
292	783–772	3.9370	7.1250	1.8750	1.8900	1.5000	0.14	0.13	4.65	4.37	6.34	6.61	1.51	14,900	9,850	-0.32
293	52393–52618	3.9375	6.1875	1.4375	1.4219	1.0313	0.14	0.13	4.57	4.29	5.59	5.98	1.23	9,900	8,050	-0.01
146	52400–52618	4.0000	6.1875	1.4375	1.4219	1.0313	0.14	0.13	4.61	4.37	5.59	5.98	1.23	9,900	[illegible],050	-0.01

For NOTES see sheets 11 and 12.

FED. SUP CLASS
3110

APPROVED 30 April 1959 REVISED Ⓑ For changes see sheets 1 thru 13

P. A. OS
Other Cust AT
11

INTERNATIONAL INTEREST

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

MILITARY STANDARD
MS19081

PROCUREMENT SPECIFICATION
FF-B-187

SUPERSEDES:

SHEET 9 OF 13

3110-0468

DD FORM 672-1 (COORDINATED)

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REVIEWER ACTIVITIES: ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES: MC, YD

Dash No.	AFBMAⒷ bearing number (See Note 12)	A	B	C	D	E	R	Y	T	P	S	O	K factor	Basic dynamic load ratings (lb)Ⓑ		Ⓑ
	Cone–Cup	Bore	Outside dia.	Bearing width	Cone width	Cup width	Ⓑ Max. shaft fillet radius	Ⓑ Max. housing fillet radius	ⒷRecommended shoulder diameter					Radial	Thrust	Effective load center
									Shaft		Housing					
147	687–672	4.0000	6.6250	1.6250	1.6250	1.1875	0.14	0.13	4.65	4.41	5.87	6.30	1.24	11,500	9,250	-0.11
148	780–772	4.0000	7.1250	1.8750	1.8900	1.5000	0.14	0.13	4.69	4.45	6.34	6.61	1.51	14,900	9,850	-0.32
149	861–854	4.0000	7.5000	2.2500	2.2650	1.7500	0.31	0.13	5.08	4.49	6.69	6.85	1.74	19,300	11,100	-0.60
294	HH221449–HH221410	4.0000	7.5000	2.2500	2.2650	1.8125	0.31	0.13	5.16	4.56	6.73	7.05	1.74	22,000	12,600	-0.59
•150	HH221449–HH221416	4.0000	7.8750	2.4375	2.2650	2.0000	0.31	0.13	5.16	4.56	6.87	7.05	1.74	22,000	12,600	-0.59
295	941–932	4.0000	8.3750	2.6250	2.6250	2.1250	0.28	0.13	5.12	4.61	7.36	7.60	1.79	24,300	13,600	-0.78
151	HH224335–HH224310	4.0000	8.3750	2.6250	2.6250	2.1250	0.28	0.13	5.20	4.76	7.56	7.94	1.79	28,300	15,800	-0.74
296	782–772	4.1250	7.1250	1.8750	1.8900	1.5000	0.14	0.13	4.80	4.57	6.34	6.61	1.51	14,900	9,850	-0.32
297	786–772	4.1250	7.1250	1.8750	1.8900	1.5000	0.25	0.13	5.04	4.57	6.34	6.61	1.51	14,900	9,850	-0.32
298	71412–71750	4.1250	7.5000	1.8750	1.9375	1.3750	0.14	0.13	4.88	4.65	6.73	7.13	1.40	15,900	11,300	-0.26
299	L521949–L521910	4.2500	5.7500	0.8438	0.8438	0.6563	0.06	0.06	4.57	4.49	5.35	5.55	1.49	4,700	3,140	0.19
152	48190–48120	4.2500	6.3750	1.3750	1.3750	1.0625	0.14	0.13	4.80	4.57	5.75	6.14	1.16	8,700	7,500	0.15
153	56425–56650	4.2500	6.5000	1.4375	1.4375	1.0625	0.14	0.13	4.84	4.61	5.87	6.26	1.18	10,100	8,600	0.08
•154	67425–67675	4.2500	6.7500	1.3386	1.1875	0.9918	0.14	0.13	4.84	4.57	6.14	6.46	1.24	7,950	6,450	0.18
300	71425–71750	4.2500	7.5000	1.8750	1.9375	1.3750	0.14	0.13	4.96	4.72	6.73	7.13	1.40	15,900	11,300	-0.26
301	HH224340–HH224310	4.2500	6.3750	2.6250	2.6250	2.1250	0.31	0.13	5.47	4.96	7.56	7.94	1.79	28,300	15,800	-0.74
155	71437–71750	4.3750	7.5000	1.8750	1.9375	1.3750	0.14	0.13	5.08	4.84	6.73	7.13	1.40	15,900	11,300	-0.26
•156	L623149–L623110	4.5000	6.0000	0.8438	0.8438	0.6563	0.06	0.06	4.84	4.72	5.63	5.79	1.41	4,900	3,480	0.25
302	64450–64700	4.5000	7.0000	1.6250	1.6250	1.1875	0.14	0.13	5.16	4.92	6.30	6.77	1.13	12,100	10,700	0.05
157	68450–68712	4.5000	7.1250	1.3750	1.2500	1.0000	0.14	0.13	5.12	4.84	6.42	6.77	1.18	8,200	6,950	0.21
158	71450–71750	4.5000	7.5000	1.8750	1.9375	1.3750	0.14	0.13	5.20	4.92	6.73	7.13	1.40	15,900	11,300	-0.26
303	938–932	4.5000	8.3750	2.6250	2.6250	2.1250	0.28	0.13	5.55	5.04	7.36	7.60	1.79	24,300	13,600	-0.78
159	HH224346–HH224310	4.5000	8.3750	2.6250	2.6250	2.1250	0.28	0.13	5.63	5.16	7.56	7.94	1.79	28,300	15,800	-0.74
304	71453–71750	4.5310	7.5000	1.8750	1.9375	1.3750	0.14	0.13	5.24	4.96	6.73	7.13	1.40	15,900	11,300	-0.26
305	L225842–L225810	4.7500	6.6875	1.0000	1.0313	0.8125	0.06	0.06	5.16	5.08	6.30	6.46	1.76	7,000	3,990	0.10
306	795–792	4.7500	8.1250	1.8750	1.8750	1.3750	0.13	0.13	5.47	5.28	7.32	7.80	1.27	16,800	13,200	-0.07
307	95475–95925	4.7500	9.2500	2.5000	2.5000	1.9375	0.25	0.13	5.87	5.39	8.23	8.54	1.58	27,300	17,300	-0.55
160	48290–48220	5.0000	7.1875	1.5625	1.5000	1.3125	0.14	0.13	5.55	5.31	6.61	6.93	1.91	12,900	6,750	-0.22
161	67388–67322	5.0000	7.7500	1.8125	1.8125	1.5000	0.14	0.13	5.67	5.43	7.09	7.44	1.70	16,900	9,950	-0.25
308	74500–74850	5.0000	8.5000	1.8750	1.8750	1.3750	0.14	0.13	5.83	5.55	7.72	8.19	1.20	17,200	14,300	0.09

FED. SUP CLASS
3110

APPROVED 30 April 1959 REVISEDⒷ For changes see sheets 1 thru 13

P. A. OS
Other Cust AT
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INTERNATIONAL INTEREST

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF
ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

MILITARY STANDARD
MS19081

PROCUREMENT SPECIFICATION
FF-B-187

SUPERSEDES:

SHEET 10 OF 13

3110-0468

DD FORM 672-1

This standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

REVIEW ACTIVITIES ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES MC, YD

	AFBMAⒷ bearing number (See Note 12)	A	B	C	D	E	R	Y	T	P	S	O		Basic dynamic load ratings (lb)Ⓑ		≉Ⓑ
Dash No.	Cone–Cup	Bore	Outside dia.	Bearing width	Cone width	Cup width	Ⓑ Max. shaft fillet radius	Ⓑ Max. housing fillet radius	ⒷRecommended shoulder diameter Shaft		Housing		K factor	Radial	Thrust	Effective load center
162	95500–95925	5.0000	9.2500	2.5000	2.5000	1.9375	0.25	0.13	6.06	5.59	8.23	8.54	1.58	27,300	17,300	-0.55
163	HH228349–HH228310	5.0000	10.0000	3.0625	3.2500	2.4375	0.38	0.25	6.46	5.83	8.78	9.20	1.82	38,200	21,000	-0.92
•164	HH231637–HH231610	5.0000	11.3750	3.2500	3.4375	2.2500	0.53	0.25	6.85	5.91	10.04	10.38	1.83	43,000	23,400	-1.05
309	799–792	5.0625	8.1250	1.8750	1.8750	1.3750	0.13	0.13	5.75	5.51	7.32	7.80	1.27	16,800	13,200	-0.07
165	799A–792	5.1250	8.1250	1.8750	1.8750	1.3750	0.14	0.13	5.83	5.59	7.32	7.80	1.27	16,800	13,200	-0.07
310	48385–48320	5.2500	7.5000	1.5625	1.5625	1.3125	0.14	0.13	5.83	5.59	6.97	7.24	1.82	13,700	7,500	-0.16
166	95525–95925	5.2500	9.2500	2.5000	2.5000	1.9375	0.38	0.13	6.54	5.83	8.23	8.54	1.58	27,300	17,300	-0.55
167	48393–48320	5.3750	7.5000	1.5625	1.5625	1.3125	0.14	0.13	5.94	5.67	6.97	7.24	1.82	13,700	7,500	-0.16
311	74537–74850	5.3750	8.5000	1.8750	1.8750	1.3750	0.14	0.13	6.10	5.83	7.72	8.19	1.20	17,200	14,300	0.09
168	74550–74850	5.5000	8.5000	1.8750	1.8750	1.3750	0.14	0.13	6.22	5.94	7.72	8.19	1.20	17,200	14,300	0.09
312	898–892	5.5000	9.0000	2.2500	2.2500	1.7500	0.14	0.13	6.30	6.02	8.07	8.50	1.39	23,000	16,500	-0.23
313	82550–82950	5.5000	9.5000	2.2500	2.2300	1.7500	0.14	0.13	6.34	6.06	8.46	8.90	1.32	23,000	17,400	-0.14
169	99550–99100	5.5000	10.0000	2.6250	2.6250	1.8750	0.28	0.13	6.69	6.14	8.94	9.37	1.43	29,200	20,400	-0.48
•170	HH231649–HH231610	5.5000	11.3750	3.2500	3.4375	2.2500	0.38	0.25	6.97	6.34	10.04	10.38	1.83	43,000	23,400	-1.05
•171	EE450551–EE451212	5.5000	12.1250	3.5000	3.6875	2.4375	0.38	0.27	7.09	6.61	10.59	10.82	1.79	44,000	24,600	-1.07
•172	HH234031–HH234018	5.5000	12.5000	3.5000	3.6875	2.6250	0.38	0.27	7.09	6.61	11.02	11.24	1.79	49,500	27,800	-1.04
•173	36690–36620	5.7500	7.6250	1.1250	1.1250	0.9063	0.06	0.06	6.10	6.02	7.17	7.40	1.59	9,700	6,100	0.19
314	36691–36620	5.7500	7.6250	1.1250	1.1250	0.9063	0.19	0.06	6.38	6.02	7.17	7.40	1.59	9,700	6,100	0.19
315	82576–82950	5.7500	9.5000	2.2500	2.2300	1.7500	0.14	0.13	6.54	6.30	8.46	8.90	1.32	23,000	17,400	-0.14
316	99575–99100	5.7500	10.0000	2.6250	2.6250	1.8750	0.28	0.13	6.89	6.38	8.94	9.37	1.43	29,200	20,400	-0.48
317	EE107057–107105	5.7500	10.5625	2.9375	2.9375	2.2500	0.25	0.25	6.93	6.54	9.33	9.82	1.51	34,400	22,800	-0.59
•174	HM231149–HM23110	5.8750	9.3125	2.2500	2.2300	1.7500	0.14	0.13	6.57	6.42	8.54	8.82	1.83	26,300	14,400	-0.45
318	99587–99100	5.8750	10.0000	2.6250	2.6250	1.8750	0.28	0.13	7.01	6.50	8.94	9.37	1.43	29,200	20,400	-0.48
175	M231649–M231610	6.0000	8.7500	1.8437	1.8437	1.3750	0.14	0.06	6.65	6.42	8.15	8.39	1.76	17,400	9,900	-0.23
176	99600–99100	6.0000	10.0000	2.6250	2.6250	1.8750	0.28	0.13	7.13	6.57	8.94	9.37	1.43	29,200	20,400	-0.48
•177	HH234048–HH234010	6.0000	12.1250	3.5000	3.6875	2.6250	0.38	0.27	7.52	7.05	10.87	11.24	1.79	49,500	27,800	-1.04
•178	HH234048–HH234018	6.0000	12.5000	3.5000	3.6875	2.6250	0.38	0.27	7.52	7.05	11.02	11.24	1.79	49,500	27,800	-1.04

•Inactive for new design after 20 August 1976.

••• Max shaft fillet radius (R) for dash number 185 is a special compound radius.

NOTES:

Ⓑ 1. Material: Cones, cups, and rollers: bearing quality - carburizing grade alloy or through-hardening grade alloy steel in accordance with FED-STD-66. Cage; carbon steel (one piece stamped).

APPROVED 30 April 1959 REVISED Ⓑ For changes see sheets 1 thru 13

FED. SUP CLASS 3110

MILITARY STANDARD

MS19081

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS, NORMAL ANGLE, TYPE 751 (TS)

SHEET 11 OF 13

3110-0468

P. A. OS
Other Cust AT 11

INTERNATIONAL INTEREST

SUPERSEDES:

PROCUREMENT SPECIFICATION FF-B-187

DD FORM 672-1 (COORDINATED)

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document

REVIEWER ACTIVITIES: ME, MI, AV, MU, IS, NS, SH

USER ACTIVITIES: MC, YD

Ⓑ 2. Hardness: Cones, cups and rollers: 58 to 64 Rockwell C.

3. Tolerances: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (A)**			
Size range Ⓑ		Tolerance	
Over	Incl.	Plus	Minus
0	3.0000	5	0
3.0000	6.0000	10	0

Cup diameter (B)**			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Bearing width (C)**			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0	4.0000	80	0
4.0000	6.0000	140	·100

Assembled bearing maximum radial runout		
Cup O.D (B)		Tolerance
Over	Incl.	
0	24.0	20

Ⓑ **Cone bore (A) tolerance is +8, –0 for dash numbers 9, 14, 184, 185 and 188.
Bearing width (C) tolerance is +140, –0 for dash numbers 9, 14, 184, 185 and 188.
Cup O. D (B) and cone bore (A) *dimensions for dash number 283 are maximum values; tolerance values are minus by the same magnitude as* positive table-values.

4. Dimensions: All dimensions are in inches. Dimensions T, P, S, and O are recommended shaft and housing shoulder diameters.
Ⓑ Dimensions R and Y are the maximum fillet radii on the shaft and the housing respectively, which will be cleared by the bearing corners.

Ⓑ 5. Effective load center: Dimension (a) locates a point on the cone axis which is the center of pressure of all resisting forces set up by the bearing rollers. All moments should be calculated from this point when determining bearing loading and shaft stresses. A minus value of (a) indicates that the center is inside the cone backface.

6. Operating temperature: Recommended operating temperature not to exceed 121° Celsius (250° Fahrenheit).

Ⓑ 7. Lubrication: Bearings shall be furnished without lubrication.

Ⓑ 8. Basic dynamic load rating: Basic dynamic load rating is that constant stationary load which a group of apparently identical bearings with stationary cups (outer rings) can endure for a rating life of 90 million revolutions of the cone (inner ring). The basic dynamic load ratings listed herein are based on a rated life of 90 million revolutions or 3000 hours at 500 rpm.

Ⓑ 9. Rating life (hours): Rating life is the number of hours at some constant speed of the cone (inner ring) that 90 percent of a group of apparently identical bearings will complete or exceed before first evidence of fatigue develops. The magnitude of the rated life in hours is found from the following:

$$L_{10} = \frac{1.5 \times 10^6}{R}\left(\frac{C}{P}\right)^{10/3} \text{hours}$$

where:

C = Basic dynamic load rating, lb.
P = Equivalent load (combined radial and thrust load), lb.
R = Revolutions per minute.

The average life is approximately five times the rating life.

10. K factor: The K factor is the ratio of the basic radial dynamic load rating to basic thrust dynamic load rating.

11. Normal angle: A normal angle bearing has a contact angle (α) between 10 and 19 degrees. The contact angle is the angle between the line of action of the *resisting forces set up by the rollers and a plane perpendicular to the bearing axis.*

Ⓑ 12. Part number: The MS part number consists of the MS number, plus the dash number. Example: MS19081-136. The AFBMA (Anti-Friction Bearing Manufacturers Association) cup and cone numbers are for reference only and are not to be used for ordering purposes.

13. For design feature purposes, this standard takes preference over procurement documents referenced herein.

14. Referenced documents shall be of the issue in effect on the date of invitation for bid.

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INTERNATIONAL INTEREST

TITLE
BEARING, ROLLER, TAPERED, SINGLE ROW OF ROLLERS; NORMAL ANGLE, TYPE 751 (TS)

MILITARY STANDARD
MS19081

PROCUREMENT SPECIFICATION
FF-B-187

SUPERSEDES:

SHEET 12 OF 13

DD FORM 672-1 COORDINATED:

3110-0468

...ry standard is approved for use by all Departments and Agencies of the Department of Defense. Selec... for all new engineering and design applications and for repetitive use shall be made from this document

REVIS... ...IES: ME, MI, AV, MU, IS, NS, SH USER ACTIVITIES: MC, YD

Ⓑ Table II, a numerical listing of dash numbers cross-referenced to bore size A, is included as information to assist in locating a known dash number in Table I.

Table II

Dash No.	Bore A	Dash No.	Bore A	Dash No.	Bore A	Dash No.	Bore A	Dash No.	Bore A	Dash No.	Bore A	Dash No.	Bore A	Dash No.	Bore A
1	0.5000	41	1.3750	81	1.9375	121	2.8750	161	5.0000	201	1.9375	241	2.2500	281	3.5000
2	0.6250	42	1.3750	82	2.0000	122	2.8750	162	5.0000	202	1.9375	242	2.2500	282	3.5000
3	0.6250	43	1.3750	83	2.0000	123	2.8750	163	5.0000	203	1.9375	243	2.2500	283	3.5433
4	0.6250	44	1.4375	84	2.0000	124	3.0000	164	5.0000	204	2.0000	244	3.2500	284	3.6250
5	0.6250	45	1.5000	85	2.0000	125	3.0000	165	5.1250	205	2.0000	245	3.2500	285	3.6875
6	0.6250	46	1.5000	86	2.0000	126	3.0000	166	5.2500	206	2.0000	246	3.2500	286	3.7500
7	0.7500	47	1.5000	87	2.0000	127	3.0000	167	5.3750	207	2.0000	247	3.2500	287	3.7500
8	0.7500	48	1.5000	88	2.0000	128	3.0625	168	5.5000	208	2.0000	248	3.2500	288	3.7500
9	0.7500	49	1.5000	89	2.0000	129	3.2500	169	5.5000	209	2.0000	No Dash No. 249		289	3.8750
10	0.7500	50	1.5000	90	2.0000	130	3.2500	170	5.5000	210	2.0000	250	2.6875	290	3.8750
11	0.7500	51	1.5000	91	2:1250	131	3.2500	171	5.5000	211	2.1250	251	2.7500	291	3.8750
12	0.7500	52	1.5000	92	2.1250	132	3.2500	172	5.5000	212	2.1250	252	2.7500	292	3.9370
13	0.8125	53	1.5000	93	2.2500	133	3.2500	173	5.7500	213	2.1250	253	2.7500	293	3.9375
14	0.8437	54	1.5000	94	2.2500	134	3.3475	174	5.8750	214	2.1250	254	2.7500	294	4.0000
15	0.8750	55	1.5000	95	2.2500	135	3.3750	175	6.0000	215	2.1250	255	2.7500	295	4.0000
16	0.8750	56	1.5625	96	2.2500	136	3.3750	176	6.0000	216	2.1875	256	2.7540	296	4.1250
17	0.9375	57	1.5625	97	2.2500	137	3.5000	177	6.0000	217	2.1875	257	2.8125	297	4.1250
18	1.0000	58	1.6250	98	2.2500	138	3.5000	178	6.0000	218	2.2500	258	2.8125	298	4.1250
19	1.0000	59	1.6250	99	2.2500	139	3.5000	179	2.2500	219	2.2500	259	2.8125	299	4.2500
20	1.0000	60	1.6250	100	2.2500	140	3.5000	180	2.7500	220	2.3125	260	2.8750	300	4.2500
21	1.0000	61	1.6250	101	2.3750	141	3.5625	181	3.3750	221	2.3750	261	2.8750	301	4.2500
22	1.0000	62	1.6250	102	2.3750	142	3.7500	182	3.5000	222	2.3750	262	2.8750	302	4.5000
23	1.1250	63	1.6250	103	2.5000	143	3.7500	183	0.7500	223	2.5000	263	2.9375	303	4.5000
24	1.1250	64	1.6250	104	2.5000	144	3.7500	184	1.0000	224	2.5000	264	3.0000	304	4.5310
25	1.1250	65	1.6250	105	2.5000	145	3.7500	185	1.2500	225	2.5000	265	3.0000	305	4.7500
26	1.1250	66	1.7500	106	2.5000	146	4.0000	186	1.2500	226	2.5576	266	3.0000	306	4.7500
27	1.1875	67	1.7500	107	2.5000	147	4.0000	187	1.2500	227	2.5625	267	3.0000	307	4.7500
28	1.2500	68	1.7500	108	2.5000	148	4.0000	188	1.3750	228	2.6250	268	3.0000	308	5.0000
29	1.2500	69	1.7500	109	2.5000	149	4.0000	189	1.4375	229	2.6250	269	3.0000	309	5.0625
30	1.2500	70	1.7500	110	2.5000	150	4.0000	190	1.4375	230	2.6250	270	3.0625	310	5.2500
31	1.2500	71	1.7500	111	2.5000	151	4.0000	191	1.5000	231	2.6250	271	3.1250	311	5.3750
32	1.2500	72	1.7500	112	2.6250	152	4.2500	192	1.5625	232	2.6250	272	3.1875	312	5.5000
33	1.3125	73	1.7500	113	2.6250	153	4.2500	193	1.6250	233	2.6250	273	3.2813	313	5.5000
34	1.3125	74	1.7500	114	2.6875	154	4.2500	194	1.7500	234	2.6250	274	3.3750	314	5.7500
35	1.3125	75	1.7500	115	2.7500	155	4.3750	195	1.7500	235	2.6250	275	3.3750	315	5.7500
36	1.3750	76	1.7812	116	2.7500	156	4.5000	196	1.7500	236	2.6250	276	3.3750	316	5.7500
37	1.3750	77	1.7960	117	2.7500	157	4.5000	197	1.8125	237	2.6875	277	3.3750	317	5.7500
38	1.3750	78	1.8125	118	2.7500	158	4.5000	198	1.8750	238	2.6875	278	3.5000	318	5.8750
39	1.3750	79	1.8750	119	2.8125	159	4.5000	199	1.8750	239	2.6875	279	3.5000		
40	1.3750	80	1.8750	120	2.8125	160	5.0000	200	1.9060	240	2.6875	280	3.5000		

FED. SUP CLASS
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APPROVED 30 April 1959 REVISED Ⓑ For changes see sheets 1 thru 13

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PROCUREMENT SPECIFICATION
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SUPERSEDES:

SHEET 13 OF 13

DD FORM 672-1 (COORDINATED)

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